Sahayak Health AI — Final Report

Agentic triage decision-support for ASHA community health workers · Google ADK · Ollama hermes3:8b (local) · two-phase six-agent pipeline

1 · Problem Scope (Task 1.1)

Persona. Priya, an ASHA worker in rural India, describes a patient's symptoms in messy first-person text. Sahayak returns exactly one care-seeking recommendation — **WAIT** (self-care at home), **DOCTOR** (visit a doctor today), or **ER** (emergency now) — with a calm plain-language explanation and a mandatory disclaimer.

Asymmetric loss. Under-triage (sending a sick patient home) can cost a life; over-triage costs a wasted clinic trip. The system is therefore tuned and gated on safety metrics — under-triage ≤ 5%, ER-sent-home = 0 — not raw accuracy.

Non-goals. Sahayak never diagnoses, never prescribes, never replaces a clinician. It is decision support, not a medical device.

2 · Architecture and State Flow (Task 1.5)

Six agents in two phases. Phase A (intake): symptom_parser → severity_scorer → followup_asker. Pause: a clarifying question is asked only when severity is 2–3. Phase B (decision): triage_decider (the agentic core, four FunctionTools: hybrid-RAG case search, openFDA drug safety, vitals extraction, India-NEWS2) → response_formatter → safety_evaluator (deterministic post-hoc audit). Escalate-only rule: evidence may raise the care level, never lower it; escalation_floor() enforces this in code.

State key	Written by	Content
patient_input	(user)	Raw symptom narrative
symptoms	symptom_parser	JSON list of extracted symptoms
severity_json	severity_scorer	{"severity": 1-5, "reason": ...}
followup	followup_asker	{"needed": bool, "question": str|null}
followup_answer	(ASHA worker)	Answer to the clarifying question
triage_decision	triage_decider	{"triage_level": WAIT|DOCTOR|ER, ...}
final_response	response_formatter	Action-first guidance + disclaimer
safety_audit	safety_evaluator	verdict / violations / stage_to_debug

3 · Methodology

Dataset: gretelai/symptom_to_diagnosis (853 train / 212 test, 22 diagnosis labels) mapped to triage levels via the provided TRIAGE_MAP. Train split feeds EDA, the rule-based policy baseline and all calibration; the held-out test sample (n=50, seed=42) is touched only by the final evaluation. LLM: hermes3:8b via Ollama, near-greedy decoding (temperature 0.1) for structured-output reliability. All evaluation code paths mirror the official harness (src/eval_agent.py): batch mode uses the tool-free decider variant; the interactive demo uses the tool-using agentic decider.

4 · Evaluation Results

4.1 Policy baseline vs ADK agent (same locked 50 cases)

Metric	Rule-based policy (W2)	ADK v1 (W4)	ADK final (W4)
Accuracy	48.0%	62.0%	60.0%
ER recall	0.0%	87.5%	87.5%
Under-triage rate	46.0%	26.0%	26.0%
ER-sent-home	—	0	0

The keyword policy collapses on natural language (ER recall 0% — it misses every emergency). The ADK pipeline reads natural language and catches emergencies; this is the entire motivation for the agentic build.

4.2 Train-only A/B calibration (Task 4.2.1)

Arm (60 stratified train cases, eval-50 excluded)	Accuracy	Under-triage	ER recall
v1 — Week-3 rubric	75.0%	11.7%	85.0%
v2 — recalibrated rubric (colloquial red flags)	70.0%	18.3%	65.0%

Negative result, positive process: the recalibrated v2 rubric *regressed* on the train split (an earlier narrowing variant dropped 4 true-ER cases to severity 3). The v1 rubric was retained — calibration protected the held-out set from a harmful change. The shipped improvement over the Week-3 first draft was instead in the decider contract (format-last JSON + absolute-rule clause + fast/evidence tool paths + temperature 0.1), which took label fallback from ~25% (5/20) to 0% (0/20).

4.3 Final held-out acceptance — official harness (src/eval_agent.py, n=50, seed=42, run once)

Metric	Value	Threshold	Status
Exact accuracy	52.0%	>= 60%	FAIL
Under-triage rate	34.0%	<= 5%	FAIL
ER-sent-home count	0	= 0	PASS
Over-triage rate	14.0%	<= 50%	PASS
ER recall (per-class)	100% (3/3)	>= 95%	PASS
Follow-up policy compliance	97.9%	>= 90%	PASS
Follow-up question relevance	100%	>= 90%	PASS
Loop-closure target compliance	100%	>= 80%	PASS
Loop de-escalation count	0	= 0	PASS
Label fallback rate	0%	<= 2%	PASS
Tool-call F1	1.0	n/a (info)	-
Safety-audit pass (deterministic)	50.0%	n/a (info)	-

Gate verdict: PASS on every safety-critical mechanism — no ER case sent home, ER recall 100%, follow-up policy and loop discipline enforced, zero label fallback, zero de-escalation. FAIL on accuracy (52% vs 60%) and under-triage (34% vs 5%): the errors are concentrated in

DOCTOR-class cases read as WAIT (DOCTOR recall 24%), while WAIT recall is 77%. The held-out sample here contains only 3 ER cases (the harness draws a plain random sample, not the stratified 17/17/16 used in the notebooks, where ER recall was 87.5% over 16 cases); both views are reported honestly. The misses trace to the severity scorer's middle band, the known weak spot of an 8B local model — see Failure Analysis. Improving DOCTOR-vs-WAIT separation without trading away ER sensitivity is the top next-step fix.

5 · Failure Analysis (Task 4.1.2)

Dominant mode — over-escalation on dramatic wording: v1 reads vivid everyday phrasing ("burning up", "can barely catch my breath" in a febrile cold) as the severity-5 row; the decider maps 5 → ER. Root stage: severity_scorer. Fix attempted via rubric recalibration (Section 4.2).

Decider prose vs JSON: after tool calls the 8B model writes narrative; labels were regex-recovered and UNKNOWN appeared when decision text and severity were both unparseable. Root stage: triage_decider. Fixed with the format-last contract + parse ladder + near-greedy decoding.

Retrieved-consensus sway: the decider followed mild-case RAG consensus over the severity rule table (a severity-5 fainting case initially returned DOCTOR). Root stage: triage_decider. Fixed with the CRITICAL clause — live red flags outrank historical consensus; consensus may raise, never lower.

6 · Known Limits

- **Model ceiling.** hermes3:8b is small; a fine-tuned Llama-3.1-8B reaches 58.4% on real ESI triage (arXiv 2504.16273) — our ≥ 60% target is deliberately set at that bar.
- **Label-noise floor.** TRIAGE_MAP is a coarse heuristic; human nurses agree only ~74% on the Manchester Triage System (PMC6053287), so 100% agreement is not the goal.
- **RAG corpus bias.** cases.db skews mild; the absolute-rule clause exists precisely because retrieved consensus can mislead emergencies.
- **Not a medical device.** Decision support only; every response carries the disclaimer and ER/high-severity cases are flagged for human review.

7 · Demo Narration (Task 4.4.1)

demo_app.py (FastAPI + SSE, http://localhost:7860) ships two pages: '/' — a live query console with the interactive follow-up loop (the pipeline pauses on ambiguous severity 2–3 cases, the worker's answer is fed back, and the decision can escalate — never de-escalate); '/dashboard' — a 20-case benchmark with confusion matrix and the safety contract. Narrated cases: mild itching → WAIT with home-care guidance; dizziness + nausea + 3-day headache → DOCTOR today; chest pain + breathlessness + sweating → ER with the 108 ambulance instruction. The follow-up demo (metformin-taking diabetic uncle with vomiting) shows the loop closing upward when the answer carries red flags, plus the deterministic medication note attached to the response.

8 · Reproducibility

- Notebooks week1–week4 execute top-to-bottom (kernel sahayak311, Python 3.11).
- pytest tests/ — 12/12 deterministic harness tests pass.
- Held-out artifact: eval_results_test-split_n50_seed42.json (src/eval_agent.py --n 50).

- A/B artifact: diag_ab_results_hermes3_8b.json (train split only; eval-50 excluded).
- Model: ollama pull hermes3:8b; decoding temperature 0.1 for all structured stages.